Bisphenol A Epoxy Resin

What is it?

Epoxy resins are chemicals used to produce glues and plastics. Usually they require two parts – a monomer unit that becomes cross-linked when a hardener is added. For most epoxies, epichlorohydrin is mixed with bisphenol A. Once it is hardened, epoxy is less likely to produce allergy. However, there may be residual resin if curing is incomplete.

BISPHENOL A EPOXY RESIN can also be known as:

- Epoxy resin
- Diglycidyl ether of bisphenol A
- Diomethane diglycidyl ether
- 2,2-bis(4-glycidyloxyphenyl)propane
- Bisphenol A diglycidyl ether
- Bisphenol A (i.e. BPA)
- Epichlorohydrin

Some Possible Uses of BISPHENOL A EPOXY RESIN / Where it might be found:

- Adhesive, glue, and sealers
- Aircraft assembly
- Anti-corrosion (e.g. oil drilling platform)
- Appliance primer and finish
- Artist, potter, and sculptor materials
- Auto tool and dye casting
- Automobile plastic and primer
- Bikes
- Boat varnish and protective coats
- Bookbinding
- Bottle caps
- Cabinet making
- Can and drum lining
- Cement waterproofing
- Circuit board assembly
- Coil coating
- Concrete crack filler
- Construction work
- Dental bonding agent
- Electric encapsulators and motors
- Electrical condensers and devices
- Electrical insulation and tape
- Electron microscopy prep
- Eyeglass frames
- Fiber glass manufacture and repair
- Fiber optic assembly
- Film cassette
- Fishing rod manufacture
- Flooring material, glue, and sealer
- Golf clubs and balls
- Handbags and purses
- Hemodialysis equipment
- Hockey stick assembly
- "Jelly" shoes
- Laminates and plastic molding
- Metal corrosion protection
- Microscopy immersion oil
- Nail polish
- Nasal cannula and pacemaker
- Paint, printing ink
- Plastic bracelets, necklaces, panties, and shoes
- Plasticizer
- Protective coating (e.g. tent fabric, patch)
- PVC products
- Shoe making and bonded leather
- Ski, ski pole, snowboard manufacture
- Surface coating (e.g. bridges)
- Synthetic asphalt
- Tennis racket manufacture
- Textile flame retardant
- Vinyl gloves

- Wall and floor protective coating
- Water proofing
- Wind turbine rotor blade production

Other products that may cross react with BISPHENOL A EPOXY RESIN:

- Other epoxy resins
- Bisphenol F
- Butylglycidyl ether
- Diamino diphenyl methane
- Diethylenetriamine
- Diisostearamidopropyl epoxypropyl - monium chloride
- Distearyl epoxypropylmonium chloride
- Hexamethylenetetramine
- Isophorone diamine
- Triethylenetetramine
- Triglycidyl isocyanurate
- Trimethylolpropane triglycidyl ether

What should I do to avoid BISPHENOL A EPOXY RESIN?

Avoid touching the epoxy resins to your skin. Prevent surface and clothing exposure to it too. It is important to note that epoxy resins can penetrate rubber gloves and equipment. Avoid breathing fumes.

Warn your dentist you are allergic to epoxy resin.

To find a possible workplace exposure, check the Material Safety Data Sheet (MSDS). Check the ingredients of all the items you use. It may be necessary to contact the manufacturer. If possible, substitute a non-epoxy resin glue, high molecular weight epoxy adhesive, or one bag epoxy product that mixes in the package.

Do not touch your face with contaminated hands or equipment. Non-contaminated or disposable facial shields may be helpful.

To protect your hands, try:

- 4H / Silvershield Gloves (www.allerderm.com, 800 - 365 - 6868 or www.northsafety.com, 800 - 430 - 4110)
- Barrier Chemical Protective Gloves (www.ansellpro.com, 800 - 800 - 0444)

A less safe alternative is to wear multiple layers of gloves, such as those made of vinyl or PVC. Thick, long sleeved PVC gloves with several disposable PVC gloves on top work well. The outer layer of gloves must then be removed every 30 to 60 minutes. Once the longer sleeved PVC glove is contaminated, it should be discarded.

Gloves made of leather or cloth are not enough for protection. Rubber or nitrile gloves may be useful, but are less so in the presence of solvents.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

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